COLLEGE MINI PROJECT PRESENTATION

Break-Even Point Predictor & Feasibility Engine

A Full-Stack Financial Decision Support Platform with Real-Time SVG Visualizations & What-If Simulations

PROJECT METADATA

Domain: FinTech / Financial Engineering & Web Systems

Developer: arjun2007arush2011-max

Architecture: Client-Side Zero-Dependency + Multi-Runtime REST

Database: Dual-Engine (SQLite3 Relational DB & LocalStorage)

Academic Scope: Mini Project for Degree Program

DEPLOYMENT & ACCESS LINKS

Live Website:
https://arjun2007arush2011-max.github.io/break_even_point/

GitHub Code:
https://github.com/arjun2007arush2011-max/break_even_point

Backend Runtimes: Python 3 standard library, Ruby WEBrick, Node.js

Hosting: Globally deployed & CDN-cached via GitHub Pages

Status: Fully functional with production test validation

BACKGROUND & MOTIVATION

Problem Statement & Project Objectives

THE PROBLEM STATEMENT

High Failure Rates: Over 65% of small business and startup failures stem from premature cost miscalculations and a lack of real-time visibility into operational break-even thresholds.

- **Static Spreadsheet Errors:** Traditional Excel sheets lack interactive dynamic graphics and are error-prone on mobile devices.
- **Unrealistic Pricing Decisions:** Entrepreneurs often underprice units without understanding how variable costs compress contribution margins.
- **Lack of Sensitivity Analysis:** Owners cannot easily simulate how a 10% inflation spike in variable costs or rent shifts required sales volume.
- **Framework Bloat:** Most commercial calculators require expensive subscriptions or heavy tracking libraries.

KEY PROJECT OBJECTIVES

✔ **Zero-Dependency Platform:** Build a self-contained web app running in vanilla ES5 with zero external JS libraries.

✔ **Inline Scalable Vector Graphics:** Render custom SVG charts with shaded profit/loss zones, break-even markers, and touch cursors.

✔ **Real-Time What-If Engine:** Provide dynamic sliders (-50% to +50%) and sensitivity matrices to stress-test cost drivers.

✔ **Multi-Runtime REST Integration:** Implement REST APIs in Python, Ruby, and Node.js with persistent SQLite3 storage.

✔ **Executive Printable Reports:** Generate professional, letterheaded feasibility reports for banks, investors, and stakeholders.

TECHNICAL DESIGN

System Architecture & Technology Stack

FRONTEND TIER

• **Vanilla ES5 JavaScript:** Universal browser support without transpilation.

• **CSS Custom Properties:** Instant light/dark theme toggle.

• **Responsive Two-Column Grid:** Adapts seamlessly across desktop and mobile screens.

• **Web Crypto API:** SHA-256 password hashing with pure JS fallback.

VISUALIZATION ENGINE

• **Custom Inline SVG:** High-performance mathematical coordinate mapping.

• **Real-Time Tracking:** Crosshair tracker with live revenue, cost, and profit tooltips.

• **Profit / Loss Polygons:** Visually shaded gradients indicating business risk.

• **Pure Math Model:** Stateless model(F, P, V, Q, T) isolating math logic.

BACKEND & DATABASE

• **Triple Runtime APIs:** Python (http.server), Ruby (WEBrick), and Node.js.

• **SQLite3 Database:** Relational tables for Users, Profiles, and Cost Inputs.

• **Debounced Sync:** Smooth server synchronization without network spam.

• **Dual-Mode Fallback:** Automatically falls back to localStorage if offline.

CORE ALGORITHMS

Mathematical Formulation & Financial Model

FINANCIAL ACCOUNTING EQUATIONS

• **Contribution Margin (CM):** CM = P - V
Per-unit revenue available to cover fixed overhead.

• **Contribution Margin Ratio:** CM% = (P - V) / P
Percentage of each dollar that contributes to paying fixed costs.

• **Break-Even Volume:** Q_BE = F / (P - V)
Sales volume required to yield exactly zero profit.

• **Break-Even Gross Revenue:** R_BE = Q_BE × P = F / CM%

• **Projected Net Profit:** Profit = Q × (P - V) - F

• **Margin of Safety:** MOS = (Q - Q_BE) / Q × 100%
Buffer percentage before business incurs an operational loss.

• **Operating Leverage:** DOL = (Q × CM) / Profit
Measures profit sensitivity per 1% change in sales volume.

ENGINE ROBUSTNESS & VALIDATION

• **Impossibility Guard (P ≤ V):**
If Selling Price is less than or equal to Variable Cost, every unit sold generates a loss. The engine flags break-even as nonexistent, shades the entire plot in red, and presents an urgent warning.

• **Operational Loss Alert (Q < Q_BE):**
If projected sales fall short of break-even, the engine computes the exact shortfall (units and dollar deficit) and highlights the loss in red.

• **Target Profit Computation:**
Computes Q_Target = (F + Target) / CM to evaluate required sales volume for desired earnings.

• **Stateless Pure Function:**
model(F, P, V, Q, T) ensures 100% test reproducibility across all client and server endpoints.

PLATFORM CAPABILITIES

Key Features & Interactive Capabilities

1. DYNAMIC INLINE SVG CHART

- Renders Total Revenue, Total Cost, and Fixed Cost lines with precision.
- Fills green-tinted Profit Zone and red-tinted Loss Zone dynamically.
- Marked Break-Even intersection with pulsing halo & coordinate badge.
- Interactive hover/touch crosshair reading real-time metrics at any volume.

2. WHAT-IF REAL-TIME SLIDERS

- Three dual-direction sliders (-50% to +50%) for Price, Var Cost, and Fixed Costs.
- Live delta indicators showing volume and dollar shifts vs baseline.
- Color-coded readouts (green for favorable shifts, red for unfavorable).
- Instant 'Reset Sliders' button to restore baseline inputs.

3. STRESS-TESTING SCENARIOS

- **Sensitivity Matrix:** Evaluates BE volume at -20%, -10%, 0%, +10%, +20%.
- **Volume Scenarios:** Pessimistic (-25%), Base, and Optimistic (+25%).
- Shows revenue, cost, net earnings, and safety margins for all scenarios.

4. BUSINESS PROFILE & PRINT REPORT

- Business profile card with 10-digit phone and email validation.
- Multi-currency selector ($, €, £, ■, ¥, CA$, A$, CHF).
- Custom volume unit labels (units, cups, boxes, meals, hours).
- Executive print layout with letterhead, KPI summaries, and charts.

CODE IMPLEMENTATION

Modular Architecture & Code Structure

FRONTEND MODULES (ES5 JAVASCRIPT)

• **model(F, P, V, Q, T):** Pure calculation engine returning break-even, profit, safety margins, and DOL.

• **AuthModule:** Tab switching, validation, client-side SHA-256 password hashing, and session management.

• **ProfileModule:** Handles business metadata, phone/email validation, and report letterhead sync.

• **MetricsModule:** Number and currency formatting, KPI card updates, and conditional alert banners.

• **ChartModule:** ViewBox coordinate projection, path generation, and hover tracking.

• **SensitivityModule:** Stress-testing matrices across driver variations (-20% to +20%).

BACKEND API & DATA PERSISTENCE

• **GET /api/health:** Service health and database status probe.

• **POST /api/auth/signup & login:** SHA-256 auth with 24-byte bearer token generation.

• **GET & POST /api/profile:** SQLite3 persistence for enterprise details.

• **GET & POST /api/inputs:** Stores cost and sales drivers with client debouncing.

• **POST /api/calculate:** Server-side calculation verification endpoint.

• **Dual-Mode Architecture:** Auto-detects server presence via /api/health. If online, syncs with SQLite3; if offline, operates via localStorage seamlessly.

CONCLUSION & IMPACT

Results, Academic Evaluation & Future Scope

PROJECT DELIVERABLES & RESULTS

✔ **Zero-Dependency Platform:** Operates without npm packages, external build tools, or CDN dependencies.

✔ **Production Deployment:** Live and accessible worldwide on GitHub Pages at https://arjun2007arush2011-max.github.io/break_even_point/

✔ **Multi-Runtime Support:** Complete, idiomatic backend implementations in Python, Ruby, and Node.js with SQLite3.

✔ **SEO & Search Engine Readiness:** robots.txt, sitemap.xml, OpenGraph cards, and descriptive metadata configured.

✔ **Real-Time Performance:** Sub-millisecond calculation updates and smooth 60fps SVG chart recalculations.

ACADEMIC EVALUATION & FUTURE SCOPE

• **Cross-Disciplinary Integration:** Combines software engineering, managerial cost accounting, relational database design, and interactive graphics.

• **Multi-Product Sales Mix:** Future expansion to calculate weighted average contribution margins for multi-item inventory catalogs.

• **Monte Carlo Risk Simulation:** Simulating demand curve fluctuations and price elasticity for probabilistic risk assessment.

• **Automated PDF Financial Dossier:** One-click generation of investor-ready feasibility binders and financial balance sheets.